Announcement

- HW Due Tuesday September 1 at 11:59PM
  - About practicing tagging problems in GradeScope
  - You must tag problems to get credit!
- If your having trouble:
  - Work with someone from class
  - Come to office hours
  - Technology Support Center (1st floor Library Wedge PL 1108, 909-537-7677).

Check yourself

Scientific notation

- Give the scientific notation with correct prefix
  - 73,600 g
  - $1.38 \times 10^3 \text{mm}/110$
- Change the prefix to scientific notation
  - Average size of human hair is ~ $70\,\mu\text{m}$
  - Average smartphone camera is 50 megapixels

Prefix	Power of 10	Abbreviation
giga-	10^9	G
mega-	10^6	M
kilo-	10^3	k
centi-	10^{-2}	c
milli-	10^{-3}	m
micro-	10^{-6}	μ
nano-	10^{-9}	n

Check yourself

Unit conversion

- The Apollo 11 space craft (first crew on the moon) traveled at ~3,000 mph what is this in m/s?
  - $1\,\mathrm{mi} = 1.609\,\mathrm{km}$
- How many seconds are in 1 year?
- The Wilshire Grand Center in downtown LA is 1,100 feet. How many miles is that? How many meters?
  - $1\,\mathrm{mi} = 5280\,\mathrm{ft}$



Check yourself

Significant digits

- Give correct sig digs: Meter stick marked with millimeter division, you measure 34.7 cm
- $31.482\ g - 24.6\ g$
- Density: $47.25\ \text{g}/5.2\ \text{cm}^3$
- 12 ball bearings have a combined mass of $38.7\ g$ what's the average mass of one?

Significant Digits Rules

- Add, subtract — keep fewest decimal places
- Multiply, divide, roots — keep fewest sig digs
- Exact numbers — never limit number of sig digs

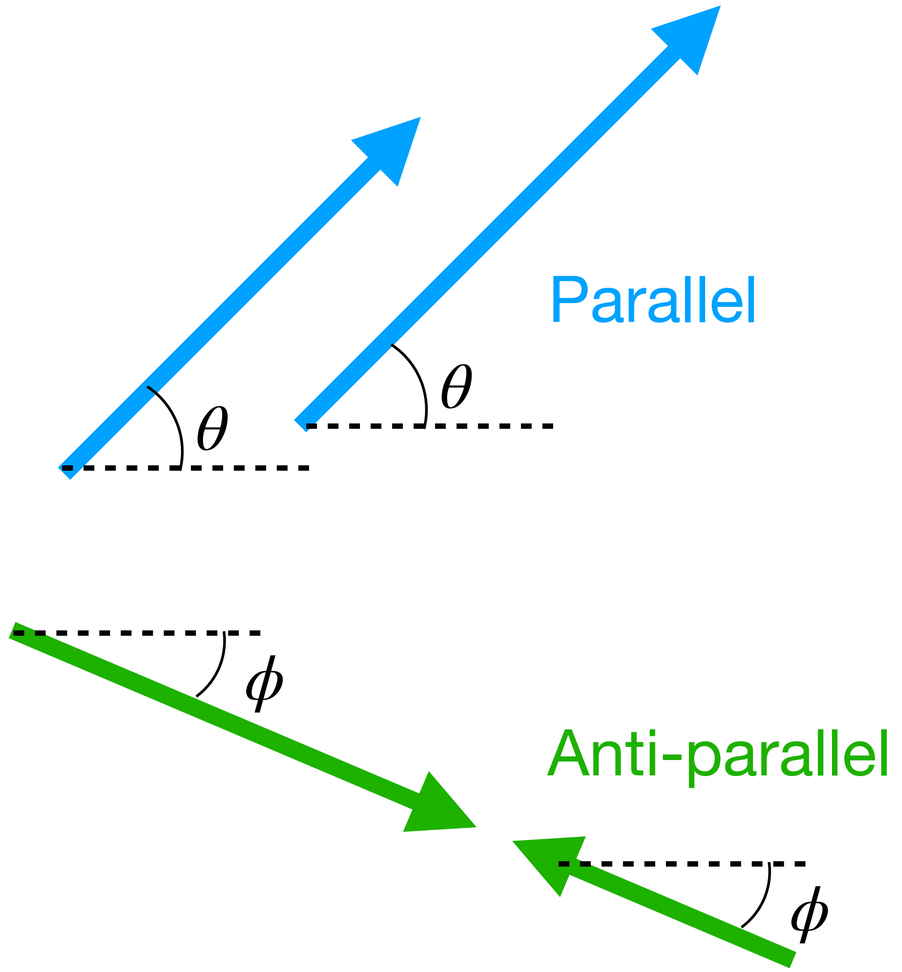
Vectors and Coordinate Systems

Two Types of Physical Quantities

- **Scalar:** fully described by a number
  - Ex: Mass, temperature, volume, speed
  - Denoted by a letter: $m = 24\,\mathrm{kg}$ (don't forget units!)
- **Vector:** described by a number and a direction
  - Ex: Velocity, acceleration, force
  - Denoted by bold letter or with an arrow: $\mathbf{v} = \vec{v} = (5\,\mathrm{m/s}, \mathrm{north})$

Vector Terms

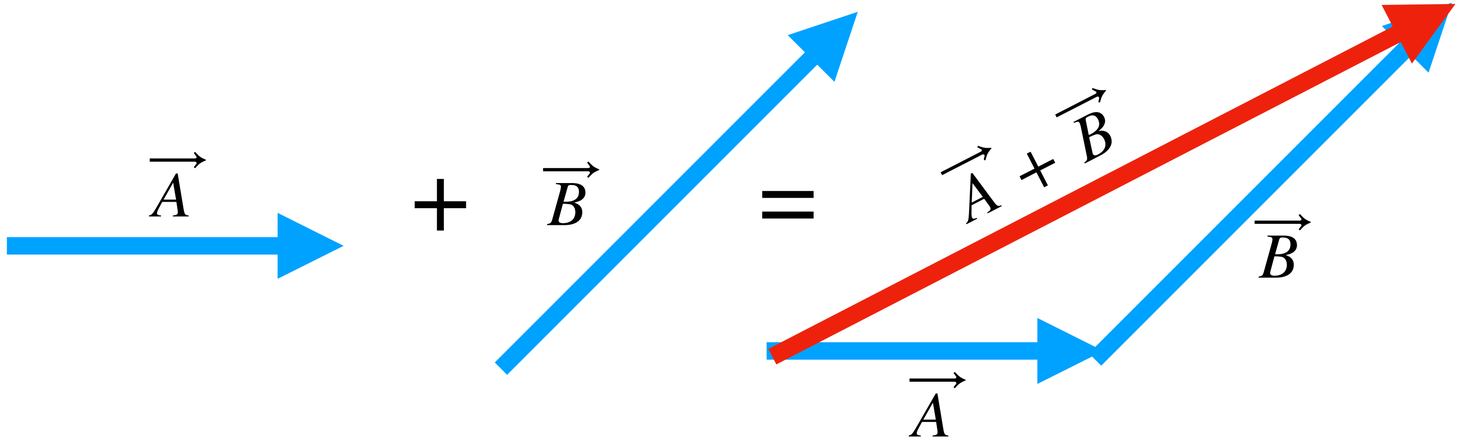
- Magnitude: length or value of a vector (the “number”)
  - Scalar
  - Denoted by $\left|\overrightarrow{A}\right|$ or A
  - Always positive
- Parallel: two vectors with same direction (not necessarily same magnitude)
- Anti-parallel: two vectors with opposite direction (not necessarily same magnitude)

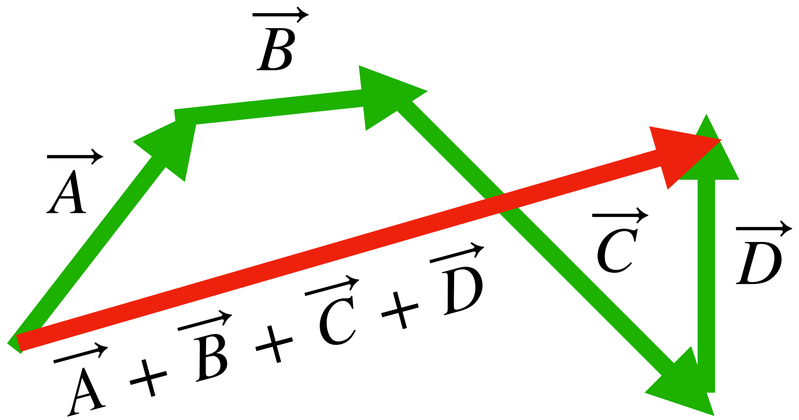


Vectors as Graphical Things

Vector operations

- Addition
  - Tip-to-tail (triangle rule)
    - Move one vector (doesn't matter which) so tail is on tip of other
      - **Don't change it! Just move it!**
    - Result vector extends from tail of first to tip of second





You try!

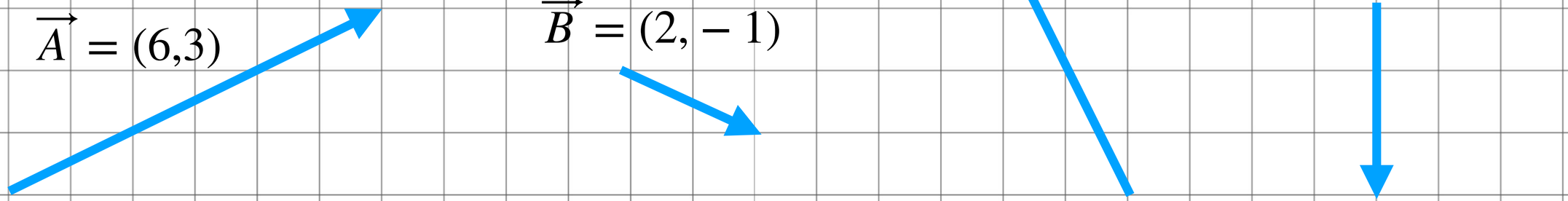
$$\vec{A} + \vec{C} + \vec{D} + \vec{B} = ?$$

$\vec{A} = (6,3)$

$\vec{B} = (2, -1)$

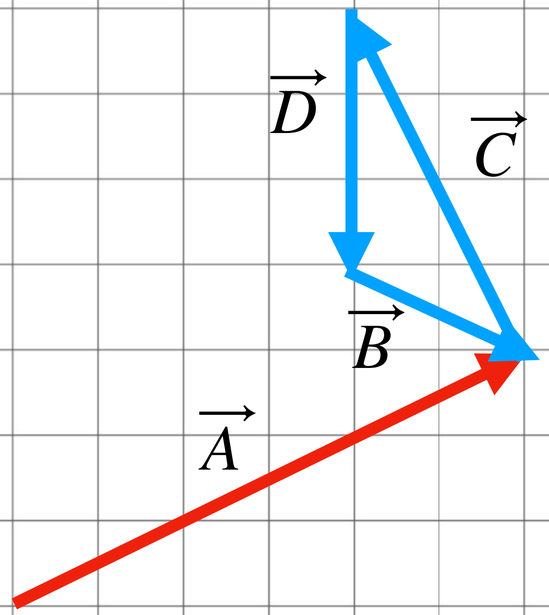
$\vec{C} = (-2,4)$

$\vec{D} = (0, -3)$



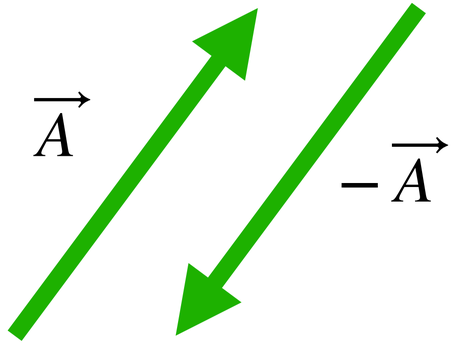
You try!

$$\vec{A} + \vec{C} + \vec{D} + \vec{B} = \vec{A}$$

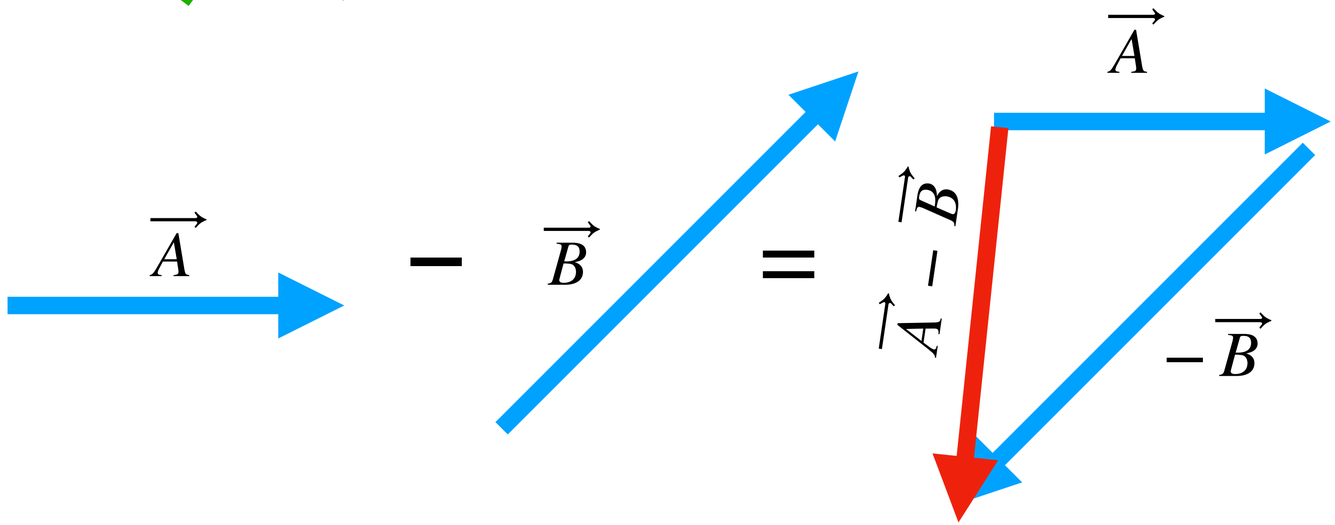


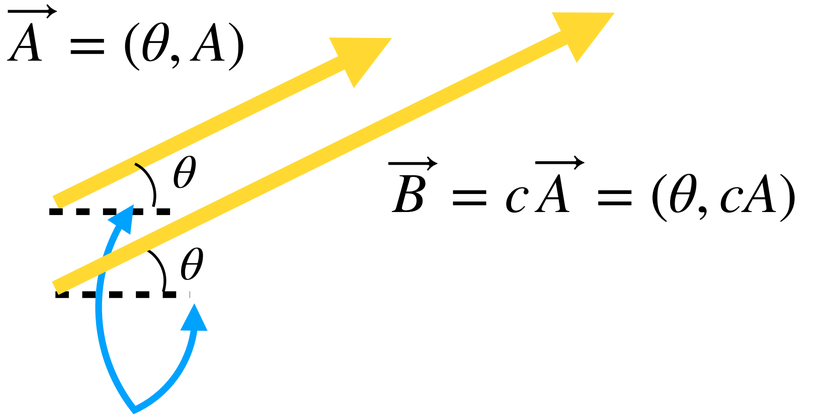
Vector Operations

- Negative vector: Same magnitude oppose direction
- Subtraction
  - Make vector negative and add
  - Order matters!
- Multiply by a scalar



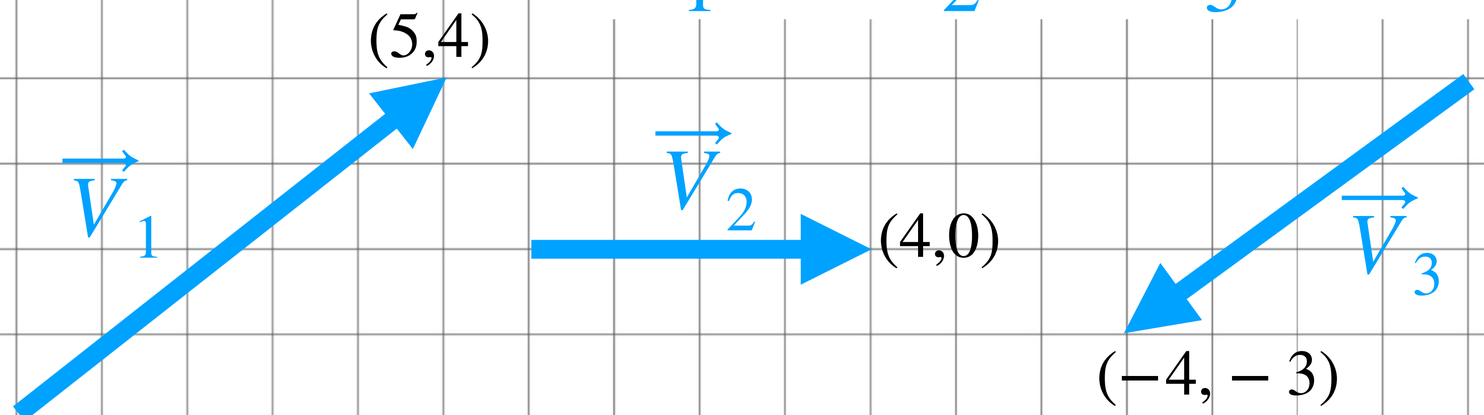
$$\vec{A} + (-\vec{A}) = 0$$





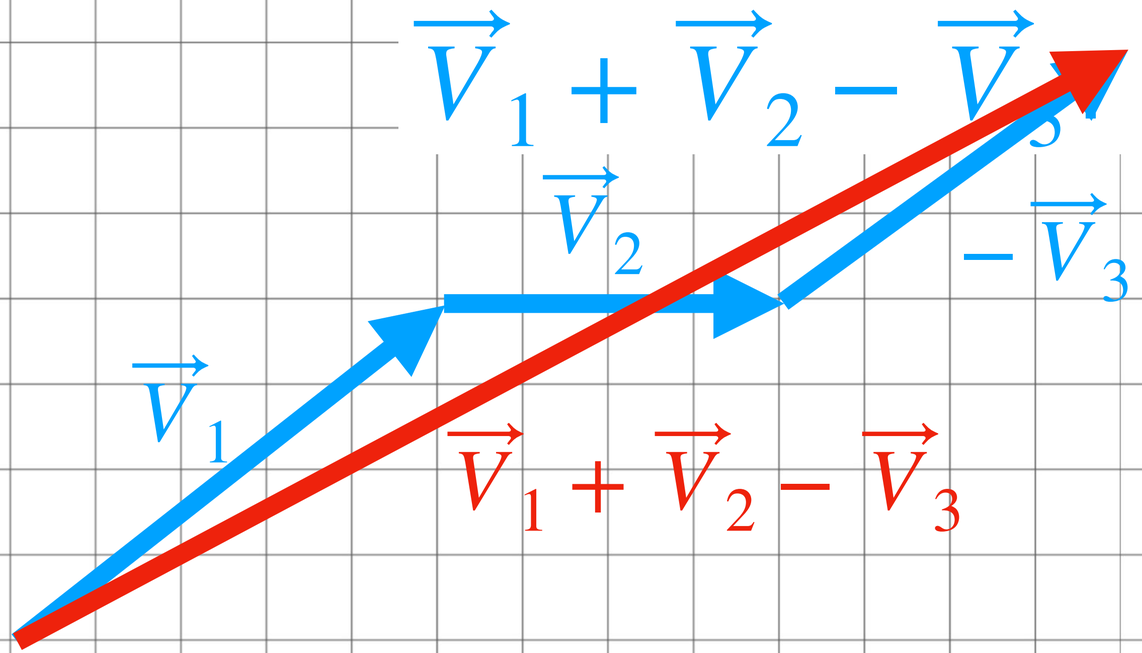
Check yourself

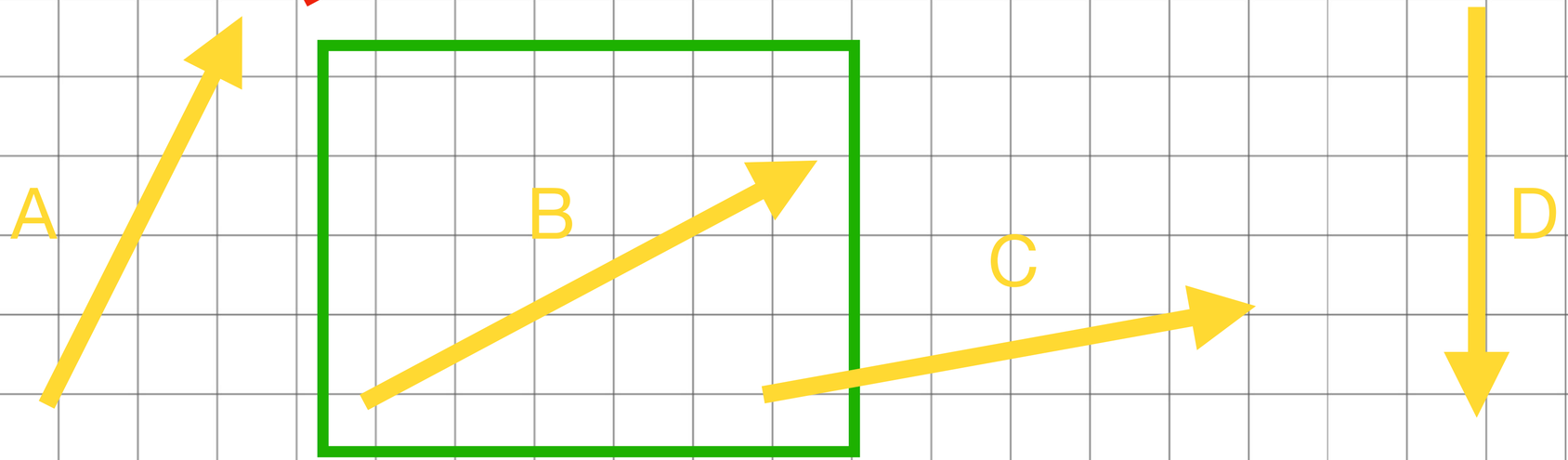
Which figure shows the direction of $\vec{V}_1 + \vec{V}_2 - \vec{V}_3$?





Which figure shows the direction of $\vec{V}_1 + \vec{V}_2 - \vec{V}_3$

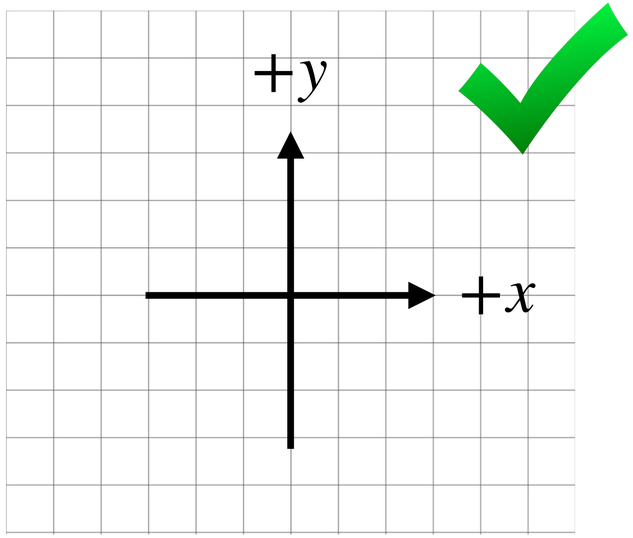


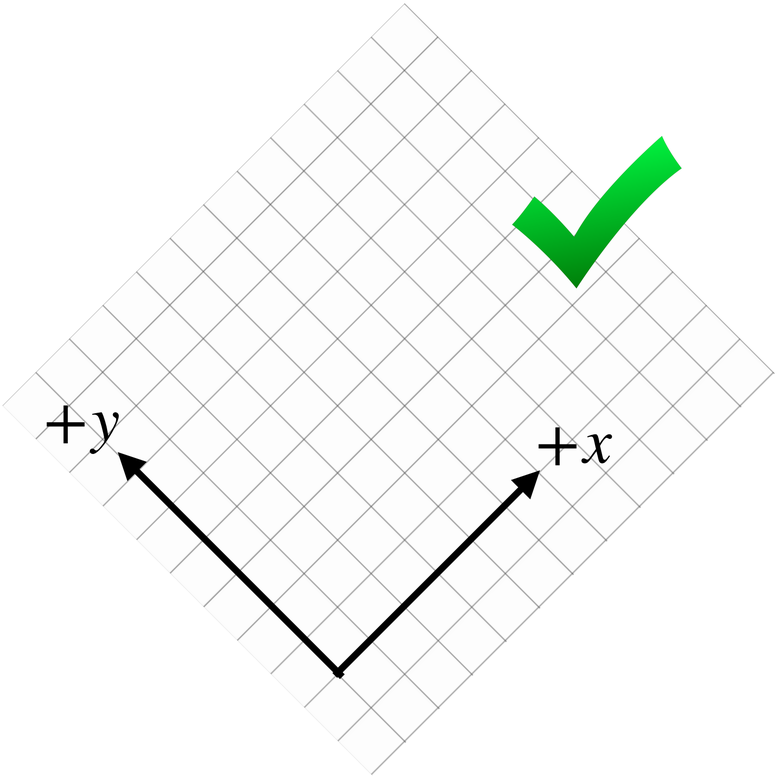


Vectors as math things

Coordinate System

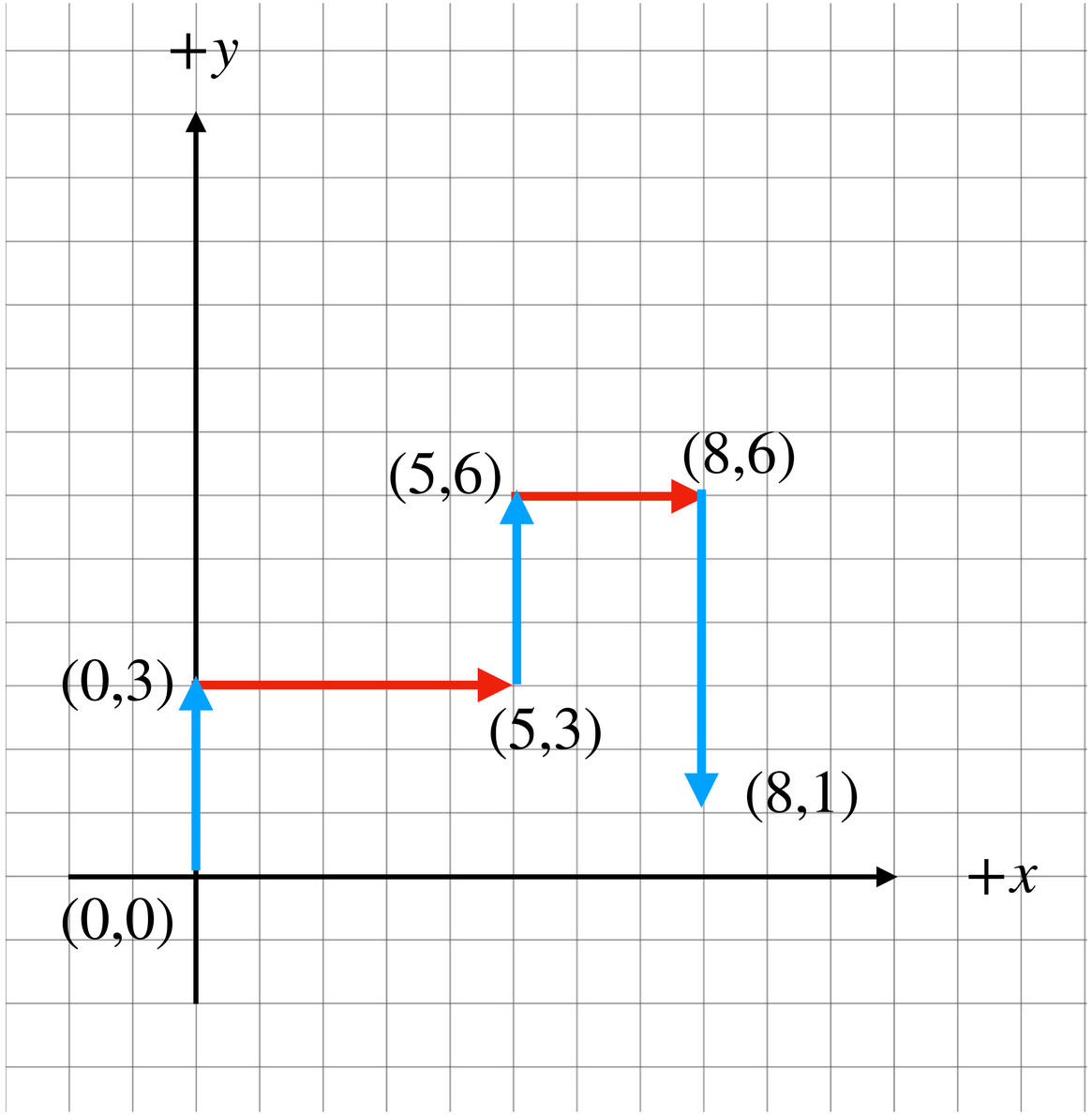
- Artificially imposed grid
- Free to choose origin and orientation of axes
- **Must label axes and +/- directions**
- We'll usually use Cartesian (xyz) coordinates





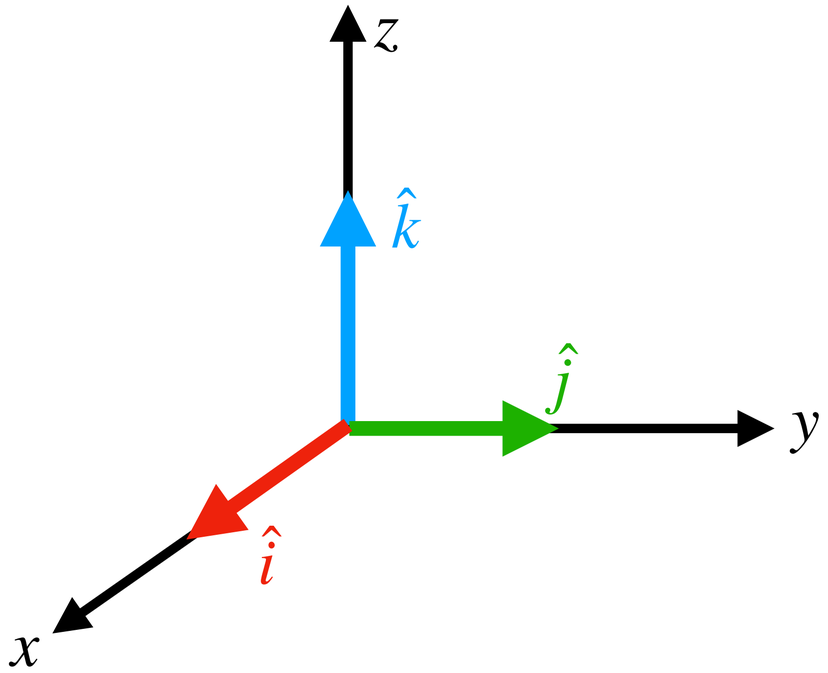
Orthogonal Coordinate Systems

- Axes are mutually independent — moving along one only changes that coordinate
- We'll only use mutually orthogonal coordinate systems (*probably*)
- Cartesian coordinates is an orthogonal coordinate system



Unit Vectors

- Any vector with magnitude 1
- Denoted with a hat: $\hat{a}$
- **Axes of coordinate systems are unit vectors**
- Denoted with: $\hat{x}, \hat{y}, \hat{z}$ or $\hat{i}, \hat{j}, \hat{k}$ or $\hat{e}_1, \hat{e}_2, \hat{e}_3$



Vector Decomposition

From magnitude and angle

- Vectors can be written like:

$$\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$$

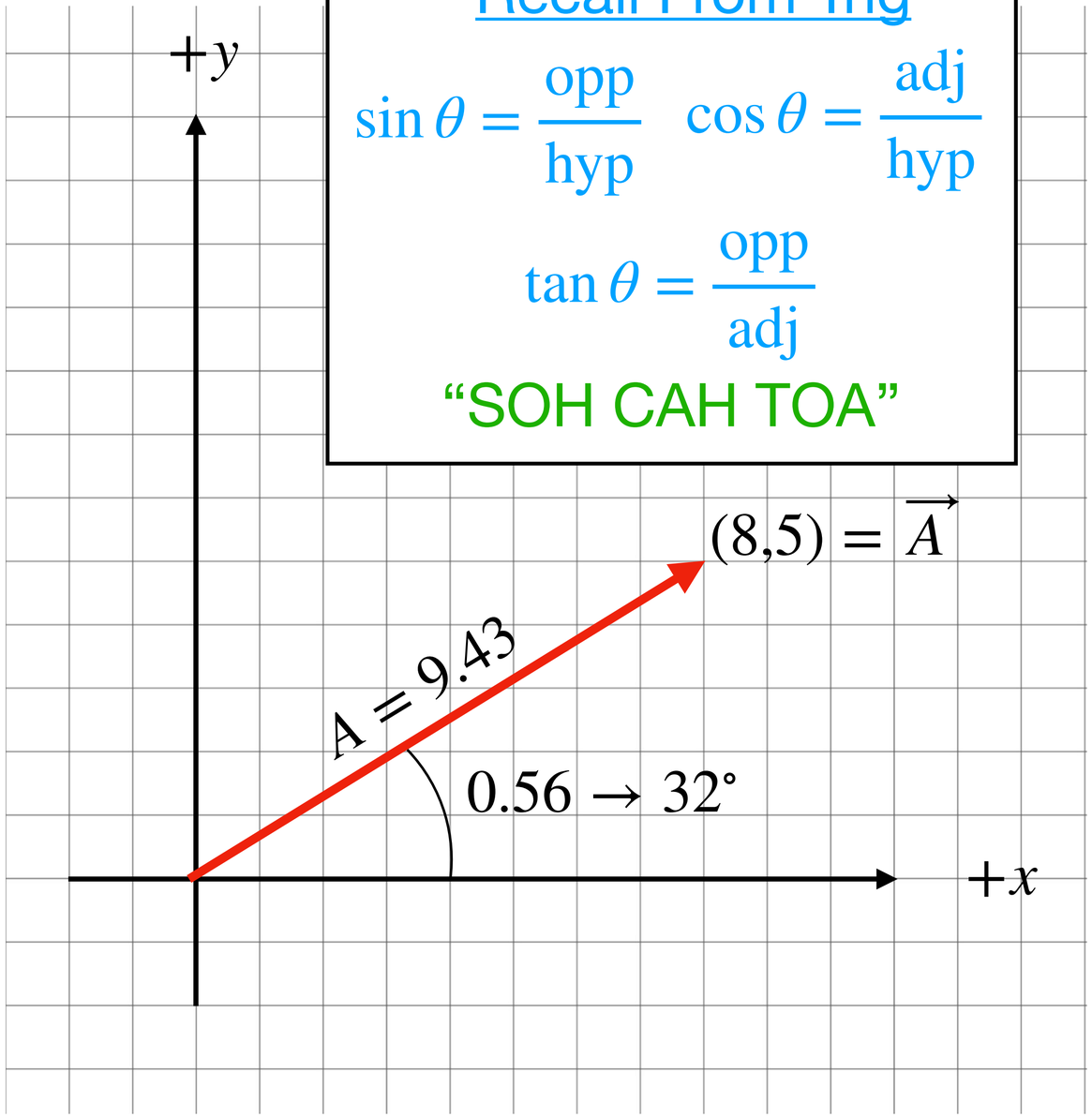
- Ex setup: “incline that rises at 30° about the horizontal box slides with a speed of $0.5\ \mathrm{m/s}$”
- Use trig
  - $A_y = A\sin(0.56)$
  - $A_x = A\cos(0.56)$
  - $\vec{A} = 8\,\hat{x} + 5\,\hat{y}$

Recall From Trig

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \quad \cos\theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}}$$

“SOH CAH TOA”



Vector Decomposition

From magnitude and angle

- Common workflow
  - Convert vectors into components
  - Do physics (in components)
  - Convert back (if asked)
- Tips:
  - Use radians (we prefer them in physics)
  - Don’t just blindly apply!
    - What if I switch the axes?

Recall From Trig

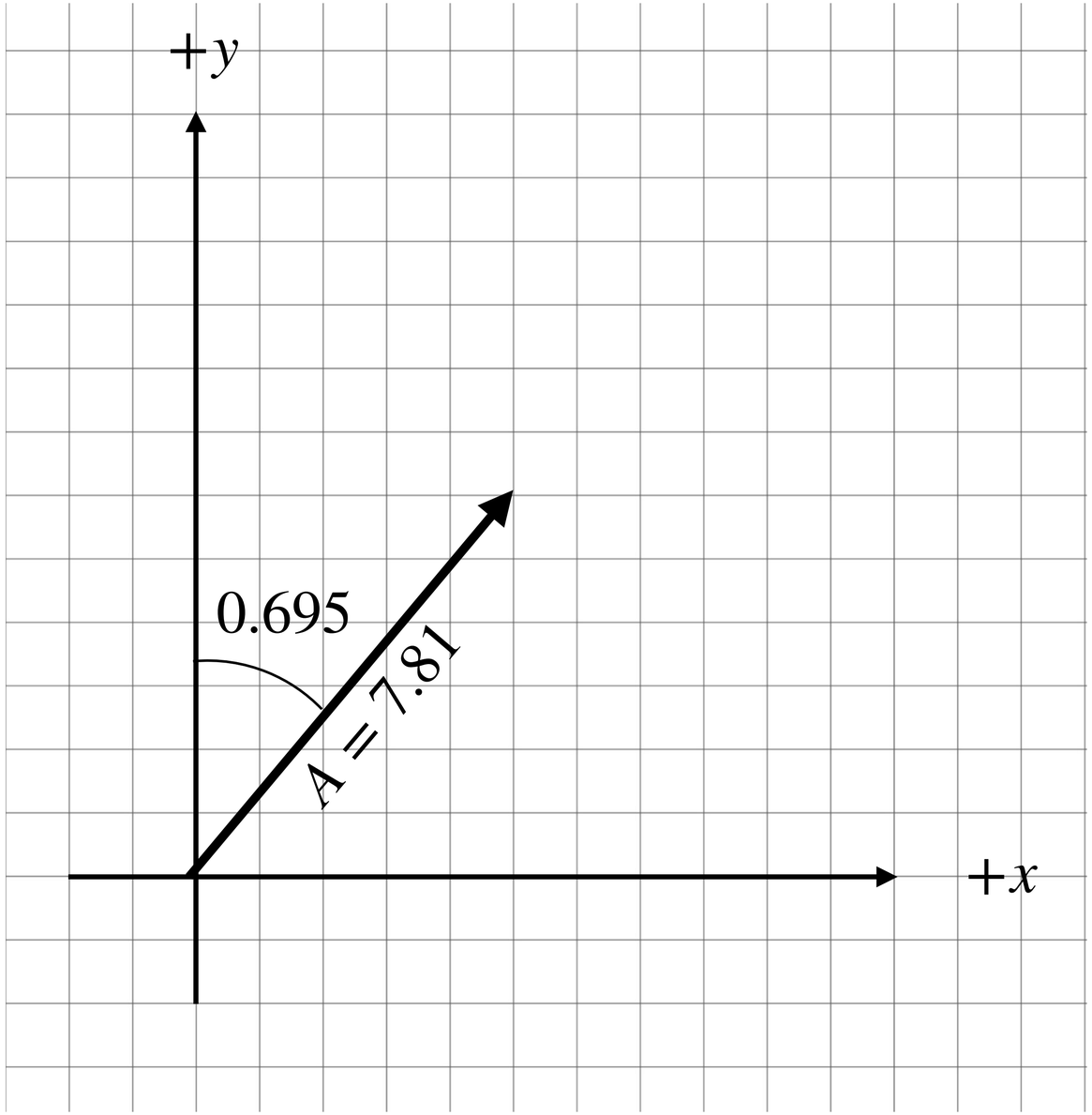
$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \quad \cos\theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}}$$

“SOH CAH TOA”

You try!

From magnitude and angle



Vector Operations (again)

- Let
  - $\overrightarrow{A} = A_x\hat{x} + A_y\hat{y}$
  - $\overrightarrow{B} = B_x\hat{x} + B_y\hat{y}$
- $\overrightarrow{A} \pm \overrightarrow{B} = (A_x \pm B_x)\hat{x} + (A_y \pm B_y)\hat{y}$
- $c\overrightarrow{A} = c(A_x\hat{x} + A_y\hat{y}) = cA_x\hat{x} + cA_y\hat{y}$

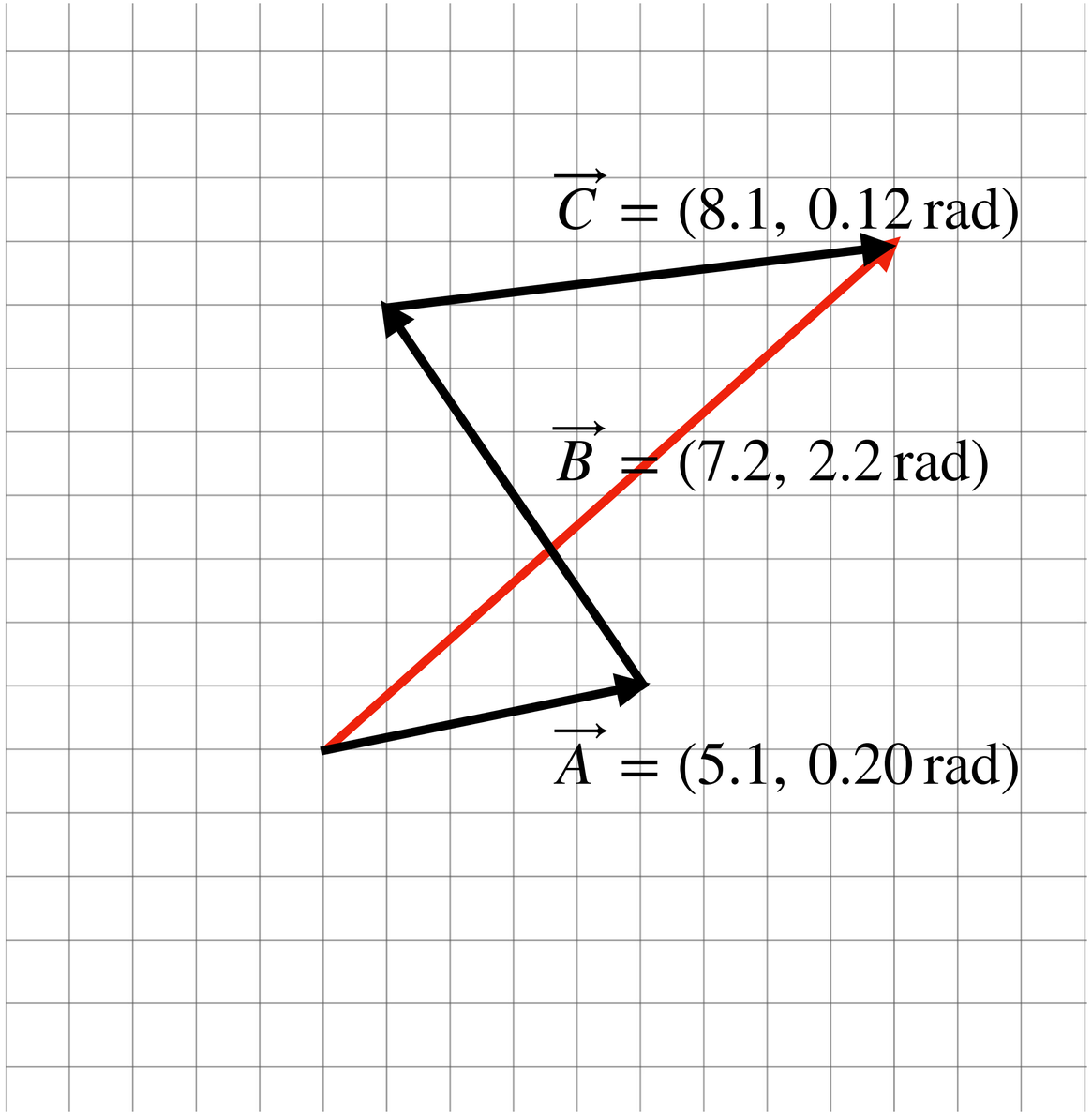
You try!

- $\overrightarrow{A} = (5.1,\ 0.20\,\text{rad})$
- $\overrightarrow{B} = (7.2,\ 2.2\,\text{rad})$
- $\overrightarrow{C} = (8.1,\ 0.12\,\text{rad})$
- $\overrightarrow{A} + \overrightarrow{B} + \overrightarrow{C} = ?$



You try!

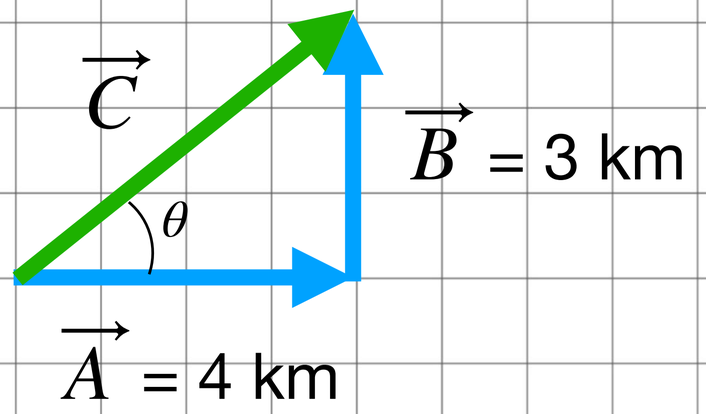
- $\overrightarrow{A} = 5.0\hat{x} + 1.0\hat{y}$
- $\overrightarrow{B} = -4.2\hat{x} + 5.8\hat{y}$
- $\overrightarrow{C} = 8.0\hat{x} + 1.0\hat{y}$
- $\overrightarrow{A} + \overrightarrow{B} + \overrightarrow{C} = 8.8\hat{x} + 7.8\hat{y}$



Components to magnitude and direction

- $\overrightarrow{A} = A_x\hat{x} + A_y\hat{y}$
- $|\overrightarrow{A}| = \sqrt{A_x^2 + A_y^2}$
- $\theta = \arctan(A_y/A_x)$
- More common to compute magnitude than angle

Example



$$\left|\vec{C}\right| = C = \sqrt{A^2 + B^2} = \sqrt{16\text{km}^2 + 9\text{km}^2} = 5\text{km}$$

$$\theta = \arctan\left(\frac{B}{A}\right) = 0.644 \text{ (or } 36.9°\text{)}$$

A note about arctan

- Plot: $\arctan(x)$ where $x \in [-\infty, \infty]$
- Horizontal asymptotes
- Calculator will only give $\phi \in (-\pi/2, \pi/2)$ i.e., quad I, IV
- Sign rule
  - Let $x = a/b$
  - if $b > 0$, quad I or IV
  - if $b < 0$, quad II or III and you need to add π to the result
  - if b=0 undefined $-\pi/2$ or $\pi/2$

